# Context Engineering

**Context Engineering** is the practice of deliberately designing, structuring, and managing what information goes into an agent's context window at each step. So the agent has exactly what it needs to reason and act well, nothing more, or nothing less.

It is the evolution of the "Prompt Engineering" instead of just a good prompt, you are architecting the entire information flow the agent sees.

Baseline, an LLM can only see what is in the context window, so the agent's intelligence is directly limited by what you put in front of it.

- **Bad Context** --> Confused/bad agent --> Wrong tool calls --> Hallucinations
- **Good Context** --> Focused agent, correct decision --> Reliable output.

## Limitation

We cannot simply put everything in the context window, but the agent tasks are long, multi-step, and involve lots of information.

**Context engineering aim to**

1. **What to include:** Not everything is relevant at every step.
   For example, a coding agent does not need the full 10000 line codebase, just the relevant file and function.
2. **What to compress or summarize:** Older conversation turns, past tool results, and completed steps can be summarized instead of kept verbatim.
3. **What to retrieve (RAG)**
   Instead of pre-loading all knowledge, pull in relevant information dynamically based on the current step.
4. **What order to put things in**:
   LLMs pay more attention to content at the beginning and end of context.